22AIE205 – Introduction to Python

Lab Assignment 7

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Lab Exercise 7.1 – Tax Calculation Portal

7.1: Develop a tax calculation portal with an option for the user to select the preferred tax regime.

Requirements:

1. Create a user-friendly GUI using Tkinter.
2. Provide input fields for the required user details, such as annual income.
3. Include an option to select the tax regime (Old Regime / New Regime).
4. Calculate the income tax based on the selected regime.
5. Display the calculation result clearly in the GUI.
6. Arrange the widgets properly using suitable Tkinter layout managers.
7. Include appropriate input validation and a reset/clear option.

Code:

IncomeTaxCalculator7_1.py > reset

```python
import tkinter as tk
from tkinter import messagebox

window = tk.Tk()
window.title("Income Tax Calculator")
window.geometry("400x500")

label1 = tk.Label(window, text="Enter Annual Income:")
label1.pack(pady=(10, 5))

annualIncome = tk.Entry(window)
annualIncome.pack()

regime_label = tk.Label(window, text="Select Tax Regime:")
regime_label.pack(pady=(15, 5))

regime = tk.StringVar()
regime.set("New Regime")
#radio buttons
old_regime = tk.Radiobutton(
    window,
    text="Old Regime",
    variable=regime,
    value="Old Regime"
)
old_regime.pack()

new_regime = tk.Radiobutton(
    window,
    text="New Regime",
    variable=regime,
    value="New Regime"
)
new_regime.pack()

def calculate_tax():
    try:
        income = int(annualIncome.get())
        if income < 0:
            messagebox.showwarning(
```

```
                "Invalid Input",
                "Income cannot be negative"
            )
            return
        selected_regime = regime.get()
        if selected_regime == "New Regime":
            if income <= 400000:
                taxAmount = 0
            elif income <= 800000:
                taxableAmount = income - 400000
                taxAmount = taxableAmount * 0.05
            elif income <= 1200000:
                taxableAmount = income - 400000
                taxAmount = (taxableAmount - 400000) * 0.10 + 400000 * 0.05
            elif income <= 1600000:
                taxableAmount = income - 400000
                taxAmount = (
                    400000 * 0.05
                    + 400000 * 0.10
                    + (taxableAmount - 800000) * 0.15
                )
            else:
                taxableAmount = income - 400000
                taxAmount = (
                    400000 * 0.05
                    + 400000 * 0.10
                    + 400000 * 0.15
                    + (taxableAmount - 1200000) * 0.20
                )
        else:
            if income <= 250000:
                taxAmount = 0
            elif income <= 500000:
                taxableAmount = income - 250000
                taxAmount = taxableAmount * 0.05
            elif income <= 1000000:
                taxableAmount = income - 250000
                taxAmount = (
                    250000 * 0.05
```

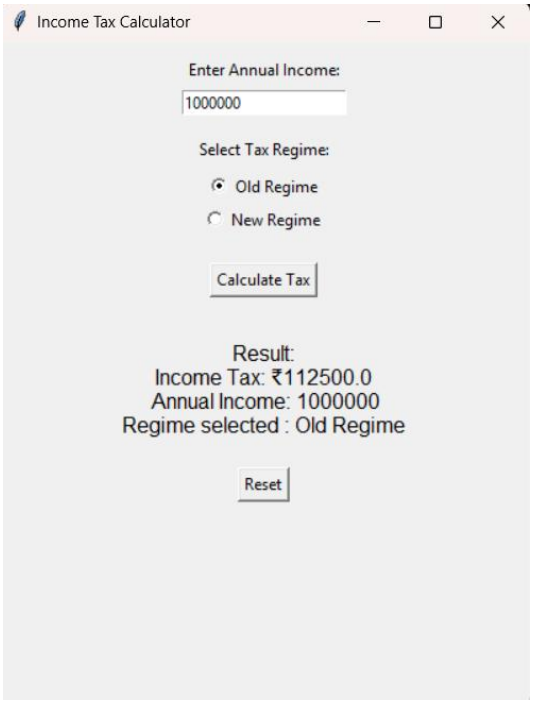
```
                )
            else:
                taxableAmount = income - 250000
                taxAmount = (
                    250000 * 0.05
                    + 500000 * 0.20
                    + (taxableAmount - 750000) * 0.30
                )

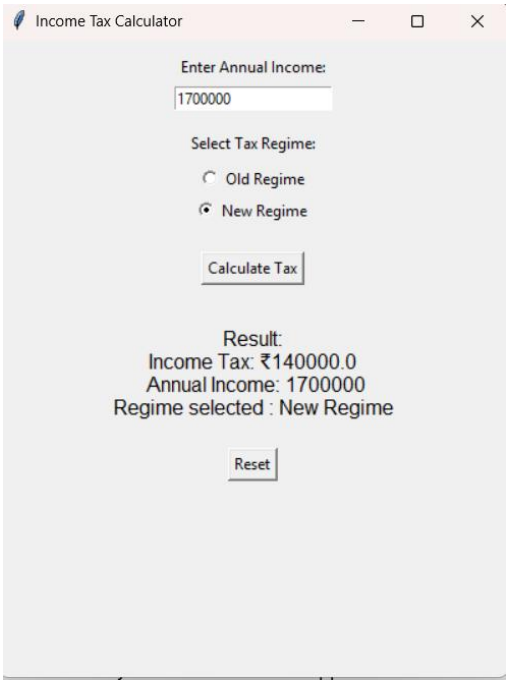
        result_label.config(
            text="Result:\nIncome Tax: ₹" + str(round(taxAmount, 2))
            +"\n Annual Income: "+annualIncome.get()
            +"\nRegime selected : "+selected_regime
        )
    except ValueError:
        messagebox.showwarning(
            "Invalid Input",
            "Please enter a valid annual income"
        )
calculate_button = tk.Button(
    window,
    text="Calculate Tax",
    command=calculate_tax
)
calculate_button.pack(pady=20)
result_label = tk.Label(
    window,
    text="",
    font=("Arial", 12)
)
result_label.pack(pady=10)
def reset():
    annualIncome.delete(0, tk.END)
    regime.set("New Regime")
    result_label.config(text="Income Tax:")
reset_button = tk.Button(
    window,
    text="Reset",
    command=reset
```

```
)
reset_button.pack(pady=10)

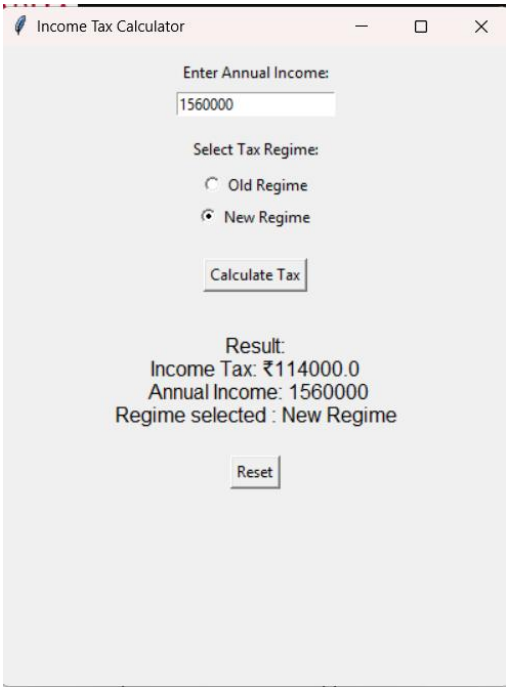
window.mainloop()
```

Multiple Validations





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Inference:

1. The Tkinter GUI successfully accepts annual income and allows the user to select the preferred tax regime.
2. The program calculates income tax according to the selected tax slabs.
3. Input validation prevents invalid or negative income values from being processed.
4. The Reset option clears the entered data and restores the default settings.

7.2: Develop a simple calculator application using Python Tkinter.

Requirements:

1. **Design a calculator GUI with a suitable window layout.**
2. **Include buttons for numbers (0–9) and arithmetic operations (+, −, ×, ÷).**
3. **Provide a display area to show the input and result.**
4. **Implement the basic arithmetic operations using button click events.**
5. **Include Clear and Equals functionality.**
6. **Arrange the widgets properly to create a neat and user-friendly design.**
7. **Handle invalid operations, such as division by zero.**

Code:

SimpleCalculatorGUI7_2.py > ...

```python
import tkinter as tk
from tkinter import messagebox

window = tk.Tk()
window.title("Simple Calculator")
window.geometry("350x450")

display = tk.Entry(
    window,
    font=("Arial", 20),
    justify="right"
)
display.grid(row=0, column=0, columnspan=4, padx=10, pady=20)

def button_click(value):
    display.insert(tk.END, value)
def clear():
    display.delete(0, tk.END)
def calculate():
    expression = display.get()
    try:
        result = eval(expression)#for evaluating math expression string
        display.delete(0, tk.END)
        display.insert(0, result)

    except ZeroDivisionError:
        messagebox.showerror(
            "Error",
            "Cannot divide by zero"
        )
        clear()
    except:
        messagebox.showerror(
            "Error",
            "Invalid operation"
        )
        clear()

buttons = [
    ("7", 1, 0),
    ("8", 1, 1),
    ("9", 1, 2),
    ("/", 1, 3),

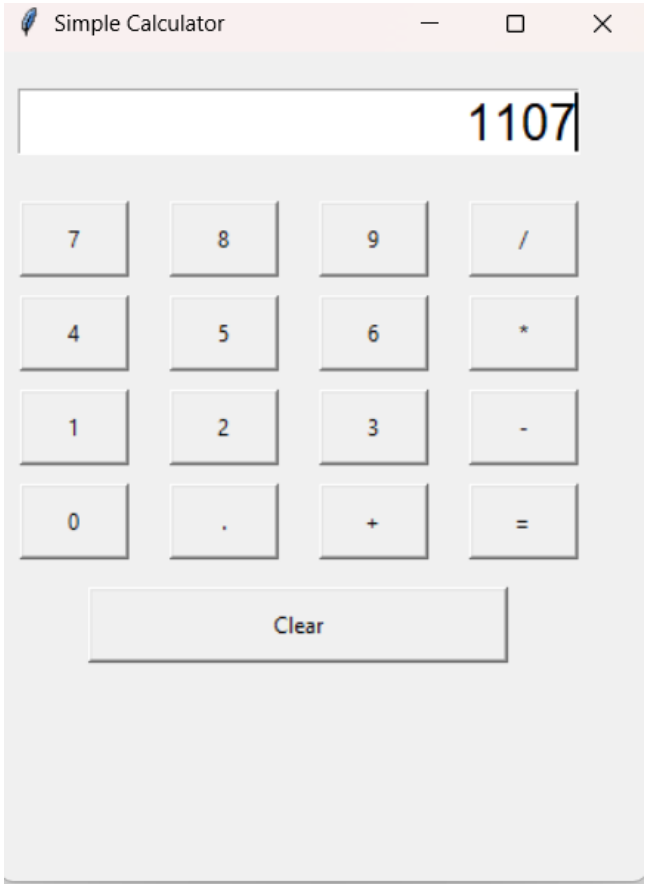
    ("4", 2, 0),
    ("5", 2, 1),
    ("6", 2, 2),
    ("*", 2, 3),

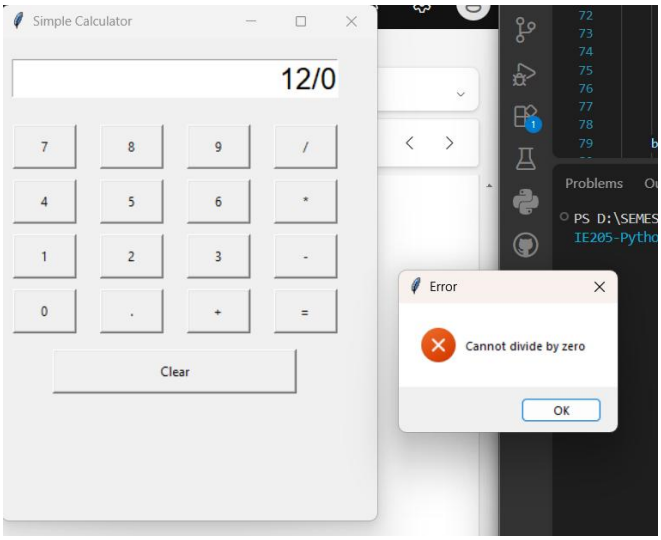
    ("1", 3, 0),
    ("2", 3, 1),
    ("3", 3, 2),
    ("-", 3, 3),

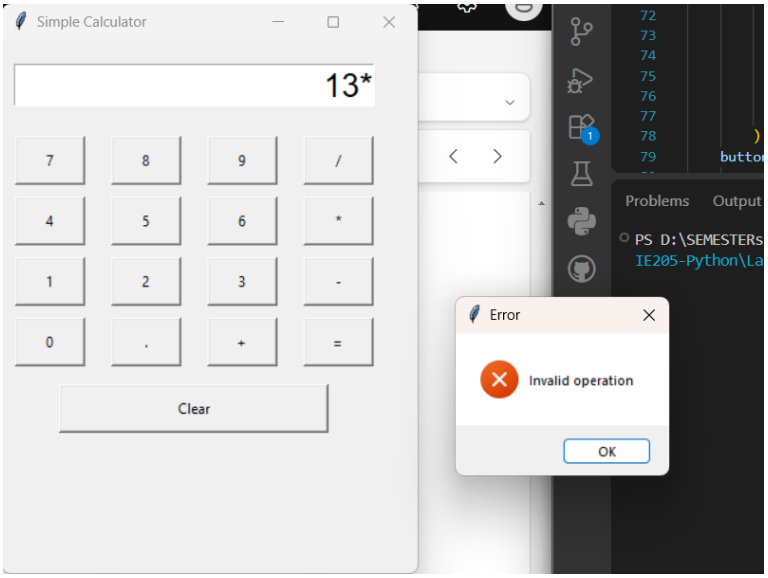
    ("0", 4, 0),
    (".", 4, 1),
    ("+", 4, 2),
    ("=", 4, 3)
]

for text, row, column in buttons:
    if text == "=":
        button = tk.Button(
            window,
            text=text,
            width=7,
            height=2,
            command=calculate
        )
    else:
        button = tk.Button(
            window,
            text=text,
            width=7,
            height=2,
            command=lambda value=text: button_click(value)
            #when this button is clciked ,send its text to button_click()
        )
    button.grid(
        row=row,
        column=column,
        padx=5,
        pady=5
    )
clear_button = tk.Button(
    window,
    text="Clear",
    width=31,
    height=2,
    command=clear
)
clear_button.grid(
    row=5,
    column=0,
    columnspan=4,
    padx=5,
    pady=10
)
window.mainloop()
```

Multiple Validations:







Inference:

1. The Tkinter calculator successfully performs basic arithmetic operations such as addition, subtraction, multiplication, and division.
2. The grid() layout manager organizes the calculator buttons in a structured row-and-column format.
3. Exception handling successfully manages invalid operations such as division by zero.
4. The Clear and Equals buttons allow the user to control the calculation and display the result.